

CS 577 — Deep Learning — Homework 1

Read these instructions carefully:

- In the L^AT_EX source code, type your answer in between “`%%% BEGIN ANSWER`” and “`%%% END ANSWER`”. For advanced L^AT_EX users, you can use your custom macros if you wish by placing them between “`%%% BEGIN MACROS`” and “`%%% END MACROS`” in the header. Do not modify anything else.
- There is a PDF file `latex_symbols.pdf` which lists L^AT_EX symbols that you will need later.
- The first section “Introduction to Latex (not graded)” is a tutorial. **It is not graded.** Feel free to skip it.
- The second section titled “L^AT_EX Problems” **will be graded.** You will recreate content from the slides in Lecture 1. You will receive full credit as long as your answer is sufficiently similar visually.
- The third section “Perceptron Problem” **will be graded.**
- When typing mathematics, you can use either the inline mode or display mode (see below). It is up to you. Please make the choice that results in good readability.
- Turn in both your `.tex` file and the generated `.pdf` file.

1 Introduction to Latex (not graded)

[0] **points** — **inline mode:** Mathematics goes inside a pair of single dollar sign like this: “`$...$`”. For example, “`$\pi \approx 3.14$`” renders as “ $\pi \approx 3.14$ ”

Write the formula for the circumference C of the circle with radius r

Answer:

[0] **points** — **display mode:** Mathematics goes inside a pair of double dollar signs for display mode like this: “`$$...$$`”. For example, “`$$\sin(\theta)^2 + \cos(\theta)^2 = 1$$`” renders as

$$\sin(\theta)^2 + \cos(\theta)^2 = 1$$

Type in display mode a function f of x and y defined (you should use colon “`:`” followed by an equal sign, i.e., `:=`, for function definition) by $\sin(x)^2$ plus $\cos(y)^2$.

Answer:

[0] **points** — **superscript:** You saw that the caret symbol `^` is for superscript. Now, try taking x to the $1/3$ power by directly typing `x^` followed by `1/3...`

Answer:

If you did this exactly, you probably noticed that it's not quite right. Let's fix that by surrounding the $1/3$ with `{...}`.

Answer:

[0] **points** — **subscript:** Subscripts in LaTeX are created using the underscore symbol `_`. For example, typing `x_i` will produce x_i . Now, try typing x with the subscript “100”.

Answer:

[0] **points** — **summation notation:** The command `\sum` is used to produce the summation symbol in LaTeX. For example, `\sum_{...}^{...}` produces $\sum_{...}^{...}$. Now, write the sum of 1 squared, 2 squared, ..., n squared using summation notation.

Answer:

[0] **points** — **blackboard font:** The set of real numbers is typically written using the blackboard font in LaTeX. To use this font, surround the symbol with `\mathbb{...}`. Now, write the set of real numbers (capital R in blackboard font).

Answer:

[0] **points** — **minimization notation:** Minimization in LaTeX is similar to summation, except there is no superscript. Now, write the notation for “minimization of x squared where x is in the set of the reals”. The symbol for “belongs” is `\in`.

Answer:

[0] **points** — **bold symbols:** To type a bold symbol in LaTeX, use the command `\mathbf{...}`, where the `...` is replaced with your symbol. Now, write X (uppercase) in bold.

Answer:

For bolding Greek symbols, you need to use `\bm{...}` instead of `\mathbf{...}`. Write “theta” in bold.

Answer:

[0] **points** — **uppercase Greek letters:** To type an uppercase Greek letter in LaTeX, simply write the name of the letter with an initial capital letter, like `\Theta` for Θ . Note that not all Greek letters have an uppercase version in LaTeX (e.g., there is no capital “alpha”). Now, type the uppercase version of delta.

Answer:

[0] **points** — **gradient symbol:** Other commands or symbols in LaTeX can also be subscripted or superscripted. The symbol for the gradient is `\nabla`. Now, type the symbol for the partial derivative with respect to bold x (just the “nabla” and the “ x ” part).

Answer:

[0] **points** — **fractions:** Fractions in LaTeX are written using the command `\frac{numerator}{denominator}`. Now, write Newton's law of universal gravitation: F is the force of gravity, G is the gravitational constant, m_1 and m_2 are the masses of the objects, and r is the distance between the objects.

Answer:

1.1 Linear algebra

[0] **points** — **matrices:** You can write matrices in LaTeX using the `\bmatrix` environment. For example, the matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is written as:

```
\begin{bmatrix}
a & b \\
c & d
\end{bmatrix}
```

Now, write down the formula for the inverse of this matrix.

Answer:

[0] **points** — **big matrices:** You can write a big matrix in LaTeX using dots for representing a sequence of elements. Here's an example of a large matrix:

$$\begin{bmatrix} x_{11} & \cdots & x_{1N} \\ \vdots & \ddots & \vdots \\ x_{N1} & \cdots & x_{NN} \end{bmatrix}$$

In this example, `\cdots` represents horizontal (centered) dots, `\vdots` represents vertical dots, and `\ddots` represents diagonal dots.

Now, create a column vector with $x_1, \dots, x_N$ and a row vector with $x_1, \dots, x_N$. Hint: You only need `&` but not `\\` for the row vector.

Answer:

1.2 Lists

[0] **points** — **nested lists:** In LaTeX, you can create a nested bulleted list using the `itemize` environment like this:

```
\begin{itemize}
  \item A
  \item B
  \begin{itemize}
    \item X
    \item Y
  \end{itemize}
\end{itemize}
```

```

        \item Z
    \end{itemize}
    \item C
\end{itemize}

```

This will produce the following nested list:

- A
- B
 - X
 - Y
 - Z
- C

To create a numbered list, use the `enumerate` environment instead of `itemize`. Nested lists can be a hacky way to write pseudo algorithms (with code indentation) in LaTeX. Now, write the pseudocode to compute the n -th Fibonacci number using nested numbered list.

Answer:

2 L^AT_EX Problems

Note: The following problems refer to slides from Lecture 01.

[2] **points** — **Empirical risk minimization:** Write the empirical risk minimization (ERM) on slide number “27/73” (ignore my handwritten notes) of . Consult the `latex_symbols.pdf` page 4 for the matrix transpose symbol.

Answer:

[2] **points** — **Gradient:** Write the gradient on slide number “29/73” (ignore my handwritten notes). Consult the `latex_symbols.pdf` page 4 for the partial derivative symbol.

Answer:

[3] **points** — **Stochastic gradient descent algorithm:** Write the content on slide number “42/73” (ignore my handwritten notes) starting from “Let...”. (Ignore the slide numbers in your answer.) The curly brackets are `\{...\}` and the left arrow is `\gets`.

Answer:

[3] **points** — **Perceptron algorithm:** Write the content inside the “Perceptron update” blue box on slide number “53/73” (ignore my handwritten notes). (Just the pseudocode. Ignore the title text of the box.)

Answer:

3 Perceptron Problem

[10] **points — Perceptron calculation:** In this problem, you will run the Perceptron update for the following dataset for $T = 8$ iterations. It is possible for the Perceptron to converge before all T iterations. In that case, you may declare that “the perceptron converges after this iteration” and stop.

- $\mathbf{x}^{(1)} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ “east”, $y^{(1)} = +1$
- $\mathbf{x}^{(2)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ “northeast”, $y^{(2)} = +1$
- $\mathbf{x}^{(3)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ “north”, $y^{(3)} = -1$
- $\mathbf{x}^{(4)} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ “northwest”, $y^{(4)} = -1$
- $\mathbf{x}^{(5)} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ “west”, $y^{(5)} = -1$
- $\mathbf{x}^{(6)} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ “southwest”, $y^{(6)} = -1$
- $\mathbf{x}^{(7)} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ “south”, $y^{(7)} = +1$
- $\mathbf{x}^{(8)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ “southeast”, $y^{(8)} = +1$

Write your answer in the box below. Show all your calculations. Go through each iteration thoroughly. Make sure that you address

- What is $\mathbf{w}$ at the beginning/end of an iteration?
- What is the calculation that goes into checking the “If/Else” statement?

Answer: